

Notes-guided MLLM Reasoning: Enhancing MLLM with Knowledge and Visual Notes for Visual Question Answering

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Abstract

*The knowledge-based visual question answering (KB-VQA) task involves using external knowledge about the image to assist reasoning. Building on the impressive performance of multimodal large language model (MLLM), recent methods have commenced leveraging MLLM as an implicit knowledge base for reasoning. However, the direct employment of MLLM with raw external knowledge might result in reasoning errors due to misdirected knowledge information. Additionally, MLLM may lack fine-grained perception of visual features, which can result in hallucinations during reasoning. To address these challenges, we propose **Notes-guided MLLM Reasoning (NoteMR)**, a novel framework that guides MLLM in better reasoning by utilizing knowledge notes and visual notes. Specifically, we initially obtain explicit knowledge from an external knowledge base. Then, this explicit knowledge, combined with images, is used to assist the MLLM in generating knowledge notes. These notes are designed to filter explicit knowledge and identify relevant internal implicit knowledge within the MLLM. We then identify highly correlated regions between the images and knowledge notes, retaining them as image notes to enhance the model’s fine-grained perception, thereby mitigating MLLM induced hallucinations. Finally, both notes are fed into the MLLM, enabling a more comprehensive understanding of the image-question pair and enhancing the model’s reasoning capabilities. Our method achieves state-of-the-art performance on the OK-VQA and A-OKVQA datasets, demonstrating its robustness and effectiveness across diverse VQA scenarios.*

1. Introduction

Visual question answering (VQA) is a task combining computer vision and natural language processing, where the

goal is to answer questions based on visual input. In recent years, there has been growing interest in VQA [1, 3, 11]. A more challenging variant, knowledge-based VQA (KB-VQA) [31, 35], requires external knowledge beyond the provided image and question to generate correct answers. These more complex application scenarios require models with enhanced capabilities to integrate vision, language, and external knowledge information. Success in this task requires not only strong multimodal understanding, but also the ability to reason with external information to produce correct answers.

Since KB-VQA relies on external knowledge to support answering, earlier approaches primarily focused on retrieving explicit information from external knowledge base to assist reasoning. RAVQA-V2 [24] utilizes Google Search to retrieve text aligned with the image and question, which is then used by an answer generator. Similarly, TRIG [9] retrieves passages from Wikidata to align visual and textual information for answer generation. With advances in MLLM performance on multimodal tasks, researchers have increasingly explored using MLLM as implicit knowledge base to improve reasoning. Prophet [37] uses GPT-3 [4], prompting it with candidate answers to improve task comprehension and generate responses. SKP [43] employs an attention mechanism to adjust the weight of each word in the explicit knowledge, which is then used directly for reasoning.

Despite the improved performance of methods based on MLLM, they exhibit several limitations:

1) The retrieved explicit knowledge may contain redundancy or incorrect information, leading to noise. If this explicit knowledge is used as input to assist MLLM in answering the question, it can distract the model and result in incorrect answers. For instance, in Figure 1(a), the question “What do they call running around...” retrieves knowledge stating, “... players trotting around the bases after hitting the ball over the fence ...” (as detailed in the Appendix Figure E1). We conducted an experiment using the image-question pair in Figure 1(a) as input, asking MLLM

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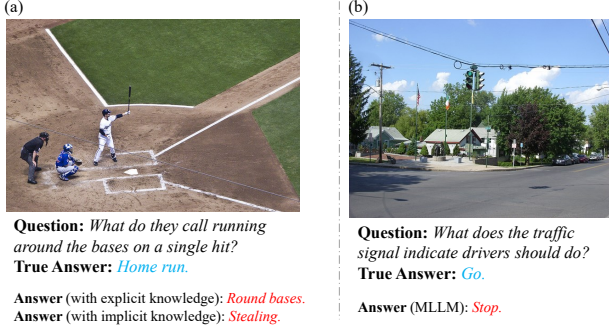


Figure 1. Limitations of the Current Approaches

to generate relevant knowledge, specific examples and analysis can be found in the Appendix A. The MLLM returned knowledge related to “Home run”, demonstrating that it possesses the correct implicit knowledge. However, when the same image-question pair was used for MLLM to directly answer the question, it incorrectly responded with “Stealing”. This suggests that while MLLM has the necessary knowledge, it fails to retrieve the correct knowledge during reasoning, leading to an incorrect answer.

2) For MLLMs, the visual encoder may struggle to capture fine-grained visual features, as noted in [6, 15], leading to hallucinations. For instance, as illustrated in Figure 1(b), when asked, “What does the traffic signal indicate drivers should do?” the model incorrectly answers “Stop” despite the image input showing a green light. This error arises from the model’s failure to detect the fine-grained visual perception in the image, resulting in an aimless and biased guess.

To address the aforementioned challenges, this paper proposes the **Notes-guided MLLM Reasoning (NoteMR)** for KB-VQA. To mitigate external knowledge noise and enhance the efficient utilization of implicit knowledge in MLLM, we propose a combined explicit and implicit approach. Input images and retrieved explicit knowledge are used as prompts, to guide the MLLM in accessing and leveraging its implicit knowledge base more effectively. This enables the model to retrieve and apply more relevant implicit knowledge for improved reasoning. We first map the question and image into a shared feature space, extract features from the external knowledge base, and retrieve the top- k most relevant entries using the mapped queries. The retrieved knowledge is then fed into the MLLM alongside the images. Leveraging the advanced visual and textual capabilities of MLLM, our approach filters explicit knowledge and guides the model to retrieve the correct implicit knowledge, generating knowledge notes that are highly relevant to the given image. To mitigate hallucinations, we further introduce visual notes that enhance the model’s fine-grained perceptual capabilities. Using the original image and gen-

erated knowledge notes, we apply a focused computation method to extract regions of the image that are highly relevant to the task. Irrelevant parts are filtered out, resulting in visual notes that focus on areas most relevant to the retrieved knowledge. Finally, the original image, question, knowledge notes, and visual notes are fed into the MLLM to produce an initial answer. To ensure consistency and reduce variance in the generated answers, the initial answers are reinjected into the model as candidate outputs, refining the output to produce the final answer.

In summary, the KB-VQA task can be likened to an exam, where our knowledge notes and visual notes serve as concise summaries of relevant information. These notes highlight key areas related to the “exam focus points” helping the model focus on the most relevant knowledge and visual content for better reasoning. Our approach offers the following contributions:

- Our approach proposes summarizing valuable knowledge notes by instructing the MLLM with retrieved explicit knowledge and images to guide its reasoning. This filters redundancy and noise in knowledge, generating more relevant information aligned with images and enhancing the model’s understanding of image-question pairs.
- Our approach proposes generating visual notes aligned with images and knowledge notes, guiding the model to focus on task-relevant visual features. This enhances the model’s fine-grained perceptual capabilities, thereby mitigating hallucination of MLLMs for KB-VQA.
- Compared with previous work, our approach achieves state-of-the-art performance. It outperforms them respectively by 5.31% on the OK-VQA dataset and by 3.4% on the A-OKVQA dataset. Experiments demonstrate the effectiveness of our approach on the KB-VQA dataset.

2. Related work

2.1. Explicit retrieval methods

The key challenge of retrieval-based methods lies in identifying relevant knowledge from external sources to answer the question accurately. Early approaches relied on fixed knowledge bases [31, 41, 45], combined with models leveraging attention mechanisms. With the introduction of OK-VQA, the focus shifted to using open-world knowledge instead of static datasets. Subsequent efforts explored various sources, including ConceptNet [27], Wikipedia [40], the Google Search Corpus [45], and more recently, GPT-3 [4], to enhance the reasoning process.

TRiG [9] utilizes image captions, labels, and OCR with text alignment to retrieve relevant knowledge passages via DPR [18]. RA-VQA [23] employs a knowledge retriever along with an answer generator to access information from the Google Search Corpus, while RA-VQA-v2 [24] extends this approach by combining text-based and image-based re-

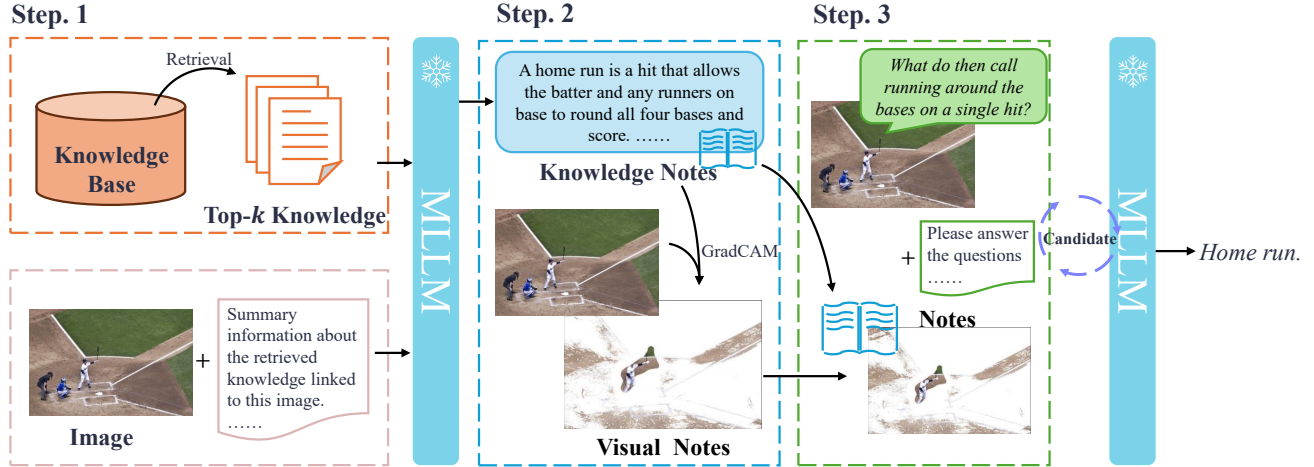


Figure 2. The framework of NoteMR

trieval, significantly improving retrieval performance. Inspired by the retrieval strength of LLM, KAT [12] and RE-VIVE [26] integrate Wikipedia and GPT-3 as additional knowledge sources, using GPT-3 outputs and DPR-based retrieval from Wikipedia to enhance answer generation. Despite improving performance, these methods often introduce noise through incorrect or redundant knowledge, which can hinder downstream reasoning.

2.2. Implicit methods with large language model

As multimodal tasks are receiving more and more attention [5, 7, 46, 50, 51], researchers are also starting to use LLM for VQA tasks. LLM based methods leverage the implicit knowledge embedded within LLMs to generate answers. For example, GPT-3 retrieves relevant knowledge to assist in answering questions. PICA [48] integrates VinVL-base [49] to generate image captions, which are then paired with questions as input to GPT-3 for answers generation. Building on PICA, subsequent works aimed to better harness the implicit knowledge of LLM. Prophet [37] first generates candidate answers from image-question pairs, using them to optimize GPT-3’s performance. PromptCap [14] constructs task-relevant captions, while ASB [47] enhances LLM performance by providing in-context examples from the training set. PnP-VQA [39] and Img2LLM [13] similarly activate LLM capabilities by generating question-specific captions via BLIP [22]. Flamingo [2] and PaLM [8] train LLMs with large samples of corpus to improve the performance of LLMs, which leads to better results. SKP [43] applies attention mechanisms on input knowledge to highlight higher-scoring words. However, this approach fails to eliminate noisy information, as emphasized knowledge remains noise if it lacks relevance.

Unlike previous methods, our approach avoids directly using retrieved knowledge to assist the MLLM. Instead, we

feed the retrieved knowledge and images into the MLLM to guide it toward relevant implicit knowledge. The resulting knowledge notes integrate both explicit and implicit knowledge, offering comprehensive information to the model. Visual notes aim to enhance the model’s fine-grained perceptual capabilities, thereby reducing hallucinations. We leverage a mechanism that links knowledge notes with image features to pinpoint key image regions relevant to the query and generate visual notes. Finally, the knowledge notes, visual notes, images, and questions are jointly fed into the MLLM to produce the answer.

3. Methodology

The framework of NoteMR is illustrated in Figure 2. First, relevant external knowledge is mined and combined with the images to generate summarized knowledge notes (Step. 1). Next, knowledge notes are utilized to identify more relevant regions within the original image, which are then retained as visual notes (Step. 2). Finally, the knowledge notes, visual notes, input images, and questions are integrated to produce the initial output. These initial outputs are then used as candidate outputs to refine the inference results (Step. 3).

3.1. Generation of knowledge notes

Unlike previous studies that focus solely on explicit retrieved knowledge or implicit knowledge in MLLMs, our approach combines both to generate summarized knowledge notes. We use explicit knowledge as prompts to guide MLLM in filtering out irrelevant explicit information and identifying the correct implicit knowledge. This synergy between explicit and implicit knowledge ensures more effective reasoning in subsequent steps. Consistent with previous work, we employ the Google Search Corpus

(GS) [45] and Wikidata [40] as external knowledge sources for OK-VQA and A-OKVQA. We extract the top- k passages related to the input image and the question with the knowledge retriever, using the pre-trained PREFLMR [25]. Specifically, we employ a text encoder $\mathcal{F}_L$ to encode the input question q and an image encoder $\mathcal{F}_V$ to encode the input image V . Following the structure of CLIP [34], we utilize $\mathcal{F}_M$ to project the image features into a shared text vector space with a hidden layer size of d_L . The final query embeddings Q are:

$$Q = [\mathcal{F}_L(q), \mathcal{F}_M(\mathcal{F}_V(V))], \quad (1)$$

where $Q \in \mathbb{R}^{l_q \times d_L}$, l_q is the length of the question q . Similarly, we can use the text encoder $\mathcal{F}_L$ to encode a document in the knowledge base to obtain its token embedding representation,

$$D = \mathcal{F}_L(d), \quad (2)$$

where $D \in \mathbb{R}^{l_d \times d_L}$, l_d is the length of the document d . Finally, the cross-correlation score r between Q and D is obtained, by the maximum dot product between the question q and the image V over all documents d , summarizing the relevance,

$$r(\bar{q}, d) = \sum_{i=1}^{l_q} \max_{j=1}^{l_d} Q_i D_j^T, \quad (3)$$

where the concatenation of the question q and the image V is denoted as $\bar{q}$. Finally, we compute the retrieval probability based on the interaction correlation between Q and D , following the methods of Lin [23] and Lewis [20].

$$p_k = p(d_k | \bar{q}) = \frac{e^{r(\bar{q}, d_k)}}{\sum_{j=1}^K e^{r(\bar{q}, d_j)}} \quad (4)$$

Ultimately, we select the top- k knowledge passages $P = \{p_1, p_2, \dots, p_k\}$ that are most relevant to the question and image.

In our study, we adopt an innovative approach of knowledge notes generation to enhance the model's understanding of external knowledge while ensuring that this knowledge is tightly connected to the information in the input images. Specifically, we introduce a pre-trained MLLM with frozen parameters to retain its ability to learn rich knowledge from multimodal data. We then provide the retrieved knowledge passages P along with the corresponding original image V as a paired multimodal input to the frozen MLLM, allowing the explicit knowledge to guide the MLLM to obtain correct implicit knowledge. Our approach leverages the MLLM to filter out noise from the knowledge passages P and extract essential information. Additionally, we utilize its deep multimodal semantic understanding to enhance the semantic correspondence between the knowledge passages and the original image, thereby improving the model's

comprehension of the input image and the accuracy of the knowledge notes. Ultimately, through the synergistic interaction of multimodal input and the frozen MLLM, our method generates comprehensive knowledge notes N_{kl} .

$$N_{kl} = \mathcal{P}_{\text{MLLM}}(c_k, V, P), \quad (5)$$

where c_k denotes the prompts that describes the knowledge notes generation task, as detailed in the Appendix Figure D1.

The summarized knowledge notes reduce noise and help extract relevant image regions. If the top- k explicit knowledge retrieved from the knowledge base is completely unrelated to the image, we return the image caption. If the explicit knowledge is related to the image, we use MLLM to summarize and filter it into knowledge notes.

3.2. Generation of visual notes

To assess the significance of each patch in relation to the image and knowledge notes, we compute the cross-modal attention matrix between the feature vector of each image patch and the knowledge notes feature vector using GradCAM [36]. To mitigate redundancy, we retain high-attention image information, guiding the language model's focus toward areas that are more pertinent to the question and knowledge, enhancing the model's fine-grained perception, and thereby alleviating the model's hallucination problem. Specifically, we divide the input image V into M patches, obtaining the image patch features $V_p^i \in \mathbb{R}^{M \times D_v^i}$. We tokenize the knowledge notes $N_{kl}^i \in \mathbb{R}^{L \times D_N^i}$ and compute the multi-head cross-modal attention scores between each patch and token.

$$head^i = \text{softmax} \left(\frac{N_{kl}^i W_q^i}{\sqrt{D_N^i}} (W_k^i V_p^i)^T \right) (W_v^i V_p^i), \quad (6)$$

where $W_q^i \in \mathbb{R}^{D_N^i \times D_N^i}$, $W_k^i \in \mathbb{R}^{D_N^i \times D_v^i}$, $W_v^i \in \mathbb{R}^{D_N^i \times D_v^i}$, denote the query, key, and value respectively of the i -th level of the multi-head attention, $head^i \in \mathbb{R}^{L \times M}$. Finally, we concatenate the multi-head cross-modal attention scores $head^i$ to obtain the final output.

$$H = \text{Concat}(head^1, head^2, \dots, head^h) W_h, \quad (7)$$

where h denotes the number of heads in the multi-head attention, and W_h represents the linear transformation matrix, with $H \in \mathbb{R}^{L \times M}$. Based on the final multi-head cross-modal attention output and the image embedding matrix, a threshold λ is set to mask out image regions with scores less than λ , while retaining the regions with scores greater than λ , thus generating a binary mask matrix.

$$\text{Mask}(i, j) = \begin{cases} 1, & H_{i,j} > \lambda \\ 0, & \text{otherwise} \end{cases}, \quad (8)$$

where $i \in [1, L]$ and $j \in [1, M]$ represent different regions, and the final binary mask matrix is denoted $\text{Mask} \in \mathbb{R}^{L \times M}$. This binary mask matrix is then element-wise multiplied with the original image to obtain the final visual notes N_{vl} .

$$N_{vl} = \sum_{i=1}^L \sum_{j=1}^M V_{i,j} \cdot \text{Mask}(i, j) \quad (9)$$

3.3. Candidate outputs

By organizing the knowledge notes N_{kl} and the visual notes N_{vl} , we represent the input as $X = \{q, V, N_{kl}, N_{vl}\}$. This allows for the simultaneous consideration of the question, relevant knowledge notes, original images, and their corresponding visual notes, thereby enhancing the accuracy and relevance of the model’s responses.

The final input for inference consists of the original image and visual notes. The visual notes highlights important regions, while the original image retains valuable contextual information from all regions.

We then input this multimodal input X into the frozen MLLM, combining it with carefully designed prompts words c (as shown in the Appendix Figure D2) to guide the model in generating an initial answer.

$$y = \mathcal{P}_{\text{MLLM}}(c, X) \quad (10)$$

We further consolidate the multiple initial answers to form a set of candidate answers. This results in a candidate answer set $Y = [y_1, y_2, \dots, y_{co}]$, co is the number of candidate answers. We then utilize the MLLM to select the best candidate from these options.

$$\bar{Y} = \mathcal{P}_{\text{MLLM}}(\bar{c}, X, Y), \quad (11)$$

where $\bar{c}$ is the prompts guide word for selecting the best candidate answer entered into the MLLM (as shown in the Appendix Figure D3).

4. Experiments

4.1. Datasets

OK-VQA [31] is a widely used knowledge-based VQA dataset comprising 9K image-question pairs for training and 5K pairs for testing. The questions are carefully filtered to ensure they require external knowledge for correct answers. Each image-question pair is annotated with ten open-ended, manually labeled answers.

A-OKVQA [35] is an enhanced version of OK-VQA, containing 17K image-question pairs for training, 1K pairs for validation, and 7K pairs for testing. The dataset offers two tasks: Direct Answer (DA) and Multiple Choice (MC). For the DA task, each question has ten open-ended, manually labeled answers, similar to OK-VQA. In the MC task,

each question provides four answer options, with one correct choice. In our work, we evaluated model performance using only the validation set.

For OK-VQA, our method utilizes the Google Search Corpus as the knowledge base, a dataset obtained through Google searches and validated in prior studies. For A-OKVQA, we use Wikipedia as the knowledge base, with information derived from Wikipedia’s web data.

4.2. Experimental setup and baselines

The text encoder in Section 3.1 utilizes the pre-trained Pre-FLMR encoder, while the image encoder is CLIP, with $k = 5$ for the top- k entries returned from the retrieved knowledge. In Section 3.2, the image encoder is BLIP, and the threshold for retaining image features is set to $\lambda = 0.6$, the number of patch sizes set to $p = 16$, the parameter M is set to 576, the parameter h is set to 12 and the candidate answers co is set to 3. Please refer to the Appendix B for specific parameter analysis. The large models employed for our reasoning are LLaVA-NeXT-7B and LLaVA-NeXT-8B. Although both models are utilized for similar durations, the 8B model demonstrates significantly better performance. Therefore, subsequent experiments focus primarily on the 8B model. Reasoning is conducted using two A6000 GPUs, requiring only 43 GB of video memory.

In our experiments, we use the standard VQA evaluation metric [31] to assess the performance of our approach. The score is calculated as follows:

$$\text{Score}(\bar{Y}, A) = \min\left(\frac{\bar{Y}}{3}, 1\right), \quad (12)$$

where $\bar{Y}$ represents the number of model-generated answers that match the ground-truth annotations in A , where A is the set of 10 manually annotated ground-truth answers for the given question. This scoring mechanism ensures a fair evaluation of partial matches.

On the OK-VQA dataset, the baseline methods listed in Table 1 are grouped into two categories:

Methods that incorporate external knowledge bases as additional inputs are called retrieval-based methods. These include ConceptBERT [10], KRISP [32], Visual Retriever-Reader [30], MAVEx [44], KAT [12], TRiG [9], RA-VQA [23], and RA-VQA-v2 [24]. Methods that rely on the internal implicit knowledge of LLMs for generating results are called LLM-based methods. These include PICA-Full [48], KAT, REVIVE [26], Flamingo [2], PromptCap [14], MM-Reasoner [19], Prophet [37], ASB [47], PaLM-E [8], SKP [43], LLaVA [28], LLaVA-NeXT [21], and Qwen2-VL [42].

On the A-OKVQA dataset, the baselines we used primarily include ClipCap [33], Pythia [16], VILBERT [29], LXMERT [38], KRISP, GPV-2 [17], PromptCap, Prophet, ASB, and SKP.

Method	Base Models	Score(%)
<i>Method based on retrieval</i>		
ConceptBERT [10]		33.7
KRISP [32]		38.4
VRR [30]		39.2
MAVEx [44]		39.4
KAT-T5 [12]	T5-large (770M)	44.3
TRiG [9]	T5-large (770M)	50.5
RA-VQA [23]	T5-large (770M)	54.5
RA-VQA-v2 [24]	T5-XL (3B)	62.1
<i>Method based on LLM</i>		
PICa-Full [48]	GPT-3 (175B)	48.0
KAT (Single) [12]	GPT-3 (175B)	53.1
KAT (Ensemble) [12]	GPT-3 (175B)	54.4
REVIVE [26]	GPT-3 (175B)	56.6
Flamingo [2]	Flamingo (80B)	57.8
PromptCap [14]	GPT-3 (175B)	60.4
MM-Reasoner [19]	GPT-4 + i-Code v2	60.4
Prophet [37]	GPT-3 (175B)	61.2
ASB [47]	LLaMA 2 (13B)	61.2
PaLM-E-12B [8]	PaLM-E (12B)	60.1
PaLM-E-66B [8]	PaLM-E (66B)	62.9
SKP [43]	Vicuna-7b	63.3
Qwen2-VL-7b [♠] [42]	Qwen2-VL-7b	58.8
LLaVA-NeXT-7b [♠] [21]	LLaVA-NeXT-7b	58.7
LLaVA-NeXT-8b [♠] [21]	LLaVA-NeXT-8b	62.2
LLaVA-13B [♠] [28]	LLaVA-13B	64.7
NoteMR	Qwen2-VL-7b	64.8
	LLaVA-NeXT-7b	68.2
	LLaVA-NeXT-8b	70.0

Table 1. Performance on the OK-VQA dataset. The best results are marked in bold. With comparable model parameters, our approach outperforms traditional retrieval knowledge methods, demonstrating superior utilization of both explicit and implicit knowledge while effectively perceiving more relevant information for accurate question answering. The [♠] notation indicates that image-question pairs are provided directly to the model as inputs for response reasoning.

4.3. Main results

To ensure a fair comparison, we employ the same evaluation metrics for both NoteMR and the baselines on the OK-VQA and A-OKVQA datasets. Additionally, we include experimental results using LLaVA directly as a base model.

The experimental results on the OK-VQA dataset are summarized in Table 1. When LLaVA is used as the baseline model, its performance improves significantly with an increase in parameter size. Notably, applying NoteMR results in substantial performance gains for both the 7B and 8B models, surpassing the performance of the 13B model.

Method	MC	DA
ClipCap [33]	44.0	18.1
Pythia [16]	49	25.2
VILBERT [29]	49.1	30.6
LXMERT [38]	51.4	30.7
KRISP [32]	51.9	33.7
GPV-2 [17]	60.3	48.6
PromptCap [14]	73.2	56.3
Prophet [37]	76.4	58.2
ASB [47]	-	58.6
SKP [43]	-	65.3
NoteMR	88.1	68.7

Table 2. Experimental results on A-OKVQA, the best results are marked in bold.

Number	Variants	Score(%)
(I)	Default	62.2
(II)	With retrieved knowledge	65.3
(III)	With knowledge notes	68.8
(IV)	(III) + visual notes	69.6
(V)	(IV) + candidate outputs	70.0

Table 3. Results of ablation experiments

This demonstrates that NoteMR without fine-tuning MLLM enables smaller-parameter models to achieve superior results compared to larger-parameter counterparts.

Although SKP integrates explicit knowledge for reasoning by directly using it, where the knowledge is filtered and fed into MLLM, it merely adjusts the weights of individual words within the retrieved knowledge and is thus unable to efficiently eliminate noise. In contrast, our approach summarizes explicit knowledge using MLLM’s implicit knowledge, and the resulting knowledge notes are then fed into MLLM for reasoning. Furthermore, it constructs visual notes that enhance the model’s fine-grained perception of images, effectively mitigating hallucinations. These two notes synergize to enhance the model’s reasoning capabilities.

To validate the effectiveness and robustness of our model, we evaluated its performance using different MLLMs as backbone networks. Additionally, we conducted experiments on the OKVQA dataset with Qwen2-VL-7b. The results demonstrate that integrating our model improved Qwen2-VL-7b’s performance by 6.1% compared to its baseline performance.

On the A-OKVQA dataset, NoteMR delivers superior performance in both Multiple Choice (MC) and Direct Answer (DA) scenarios, as shown in Table 2. In the DA scenario, which resembles the OK-VQA setup by requiring the

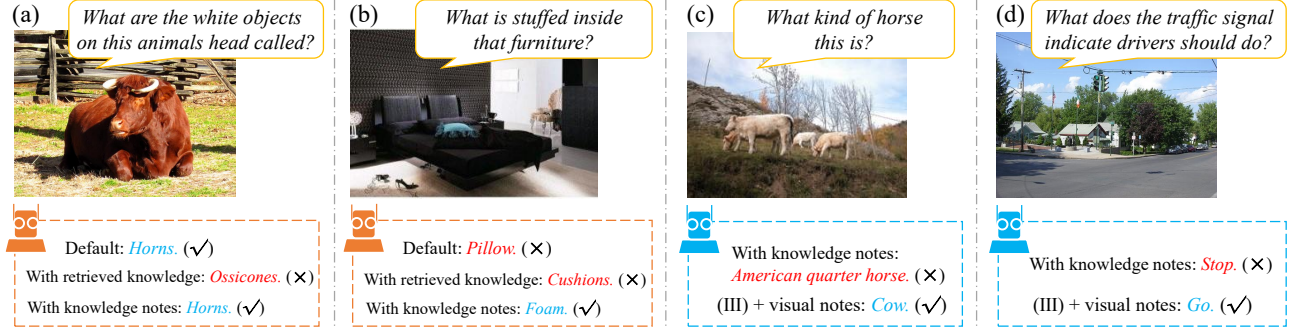


Figure 3. Case study of our method. This part of the experimental setup refers to the ablation study (Section 4.4).

model to generate answers based on the question and image, NoteMR outperforms SKP by 3.4%, showing its ability to outperform state-of-the-art methods across diverse datasets. In the MC scenario, NoteMR achieves 11.7% higher accuracy than Prophet. Since MC provides four answer options as additional prompts, it inherently enhances the reasoning of MLLM, resulting in better performance compared to DA. Prophet does not incorporate explicit knowledge to support MLLM inference, relying instead on examples similar to those in the training set to guide reasoning. Instead, we utilize knowledge notes and visual notes to guide the MLLM, effectively enhancing inference without relying on the resemblance to the training set.

These results highlight the robustness of NoteMR in diverse scenarios, achieving superior performance on multiple datasets.

4.4. Ablation study

We conducted an ablation study on the OK-VQA dataset using LLaVA-NeXT-8b as the base model. The results of the ablation experiments are shown in Table 3, below is an analysis of the results.

Default: The MLLM uses only the original images and questions as input to generate answers, achieving a score of 62.2%.

With retrieved knowledge: Incorporating raw external knowledge from the Google Search Corpus as auxiliary input improves the performance of MLLM to 65.3%, demonstrating the importance of additional information.

With knowledge notes: The incorporation of knowledge notes to assist MLLM reasoning yields a 3.5% performance improvement over directly using raw retrieved knowledge. This highlights the effectiveness of knowledge notes in filtering noise from explicit knowledge and guiding the MLLM toward relevant tacit knowledge, thereby enhancing reasoning and comprehension.

(III) + visual notes: Adding visual notes further increases the score to 69.6%. These notes focus on regions of interest in the image relevant to the knowledge and the

question, enhancing the fine-grained perception capabilities of MLLM to reducing hallucination errors. There are no candidate outputs for this experiment, y in Eq. 10 represents the final prediction.

(IV) + candidate outputs: To mitigate variability in responses, we use candidate outputs from the MLLM as reference options. This method leverages diversity in the outputs, helping the model make more informed decisions and achieve the highest score of 70.0%.

4.5. Case study

Based on our ablation study, we present a case study to illustrate the effectiveness of our approach. Figures 3 (a) and (b) compare the performance of models using external knowledge and knowledge notes, highlighting the limitations of relying solely on explicit (With retrieved knowledge) or implicit knowledge (Default) for reasoning without using knowledge notes. Figures 3 (c) and (d) evaluate the results before and after incorporating visual notes, demonstrating the deficiencies of MLLM in fine-grained perception of images without visual notes integration. (as detailed in the Appendix Figures E2 to E4)

Effects of knowledge notes. In Figure 3 (b), directly using retrieved knowledge introduces noise, leading the model to incorrectly reason and produce “Cushions” as the answer instead of the correct “Foam”. Specifically, as shown in Figure 4, for a given question q and image V , the top- k retrieved passages. These passages contain valuable information but also include significant noise. Directly feeding these passages into the MLLM risks introducing noise, such as “put batting over my cushions”, which misdirects the model and results in incorrect answers.

Moreover, MLLMs often struggle to identify and utilize the correct implicit knowledge for reasoning. As shown in Figure 3(b), relying solely on the MLLM as the implicit knowledge base results in the incorrect reasoning of “Pillow”. This occurs because the MLLM focuses on surface-level elements in the image, neglecting the more relevant knowledge that would assist in accurately answering the

Knowledge Base:

Knowledge 1:

the type of stuffing material will depend on how the upholstered furniture is to be used and the comfort level you are seeking. in cushions and other upholstery, **batting is applied over the foam to prevent slippage and provide a smooth appearance.** retardant inside the foam? what kind of material can i wrap my chair cushions with and who can do it? thanks jane reply dodie november 10, 2014 at 11:47 am do i have to **put batting over my cushions**? i washed the covers, and it appears the batting is crumpled..is that what makes my cushions look messy..??\u2026or do i just need to buy new batting.\u2026thanks a bunch reply cleaner november 30, 2014 at 9:41 am i want to recover and replace the padding on my dining room chairs.

Knowledge 2:

high-density foam provides a firmer couch cushion stuffing but may be too much for some people.

Knowledge 3:

upholstery filling whether high density cushion foam, hollowfibre filling, feather and down or a mixture of any or all is the second half of the sofa's internal construction and equally important.

Knowledge 4:

high-density foam provides a firm couch cushion but some people may find it too hard. lumpy over time unless the down is sewn into individual pockets or baffles.

Knowledge 5:

then you walk into another store and see the same fabric on a sofa that is \$500. reputation, type of cushion -- from foam to down -- and your budget.



What is stuffed inside that furniture?

Knowledge Notes:

The type of stuffing material for upholstered furniture depends on the intended use and comfort level. **Batting is applied over foam to prevent slippage and provide a smooth appearance.** **High-density foam provides a firmer couch cushion stuffing** but may be too much for some people. The type of stuffing used inside a couch varies from piece to piece, with most modern couches having foam inside the frame, seat cushions, and back cushions. Stuffing new foam into cushion covers can be done by a professional upholsterer, who will cut the foam, adhere dacron batting, stuff the new cushion insert, and steam the finished product.

Figure 4. Examples of explicit knowledge are summarized as knowledge notes. The highlighted areas represent the model’s process of filtering and summarizing explicit knowledge, where the gray area indicates relevant implicit knowledge generated by the MLLM. Red text marks noise that misleads the model into incorrect answers, while blue text marks knowledge that correctly guides the model’s reasoning.

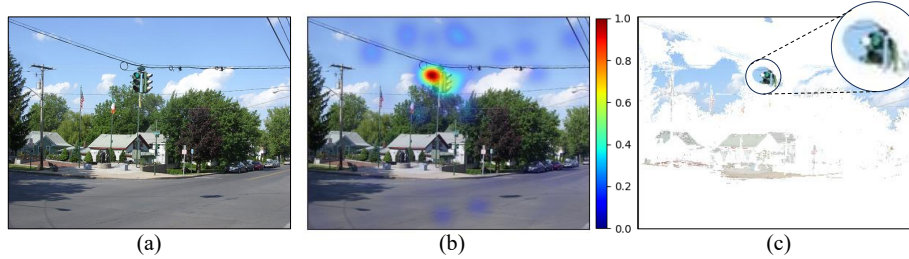


Figure 5. Examples of visual notes. (a) The original input image. (b) The attention heatmap, where red areas indicate higher relevance for answering the question. (c) The final visual notes, containing essential image features to assist the MLLM in generating correct answers.

question.

As shown in Figure 4, the knowledge notes effectively summarize explicit knowledge by filtering out noise and guiding the MLLM to retrieve relevant implicit knowledge, thus proving the practicability of knowledge notes. The blue text marks the information in the knowledge notes that assist in answering the question, enhancing the model’s understanding of the image-question pair and leading to the correct answer “Foam”.

Effects of visual notes. MLLMs are prone to hallucinations during reasoning, often arising from the insufficient ability to capture fine-grained visual details, leading to incorrect answers. To address this issue, we compute attention between the input images and knowledge notes to generate visual notes. The attention matrix focuses on image regions most relevant to answering the question, highlighting features essential for reasoning. By emphasizing these key regions, the generated visual notes guide the MLLM toward better understanding of the image. As shown in Figure 5, with the addition of visual notes, the model accurately identifies that the traffic light in the image is green, leading

it to conclude that vehicles should “Go” rather than “Stop”. This process enhances the fine-grained perceptual capabilities of MLLM, thereby alleviating hallucinations.

5. Conclusion

In this paper, we propose a methodology called NoteMR to guide better reasoning for MLLM through the utilization of knowledge notes and visual notes. Firstly, knowledge notes filter out the noise in explicit knowledge and direct the model to effectively utilize relevant implicit knowledge. By ensuring the MLLM focuses on the most pertinent information, knowledge notes significantly enhance the model’s reasoning capabilities. Secondly, visual notes enhance the fine-grained perceptual capabilities of MLLM, focusing on key image regions relevant to the task. These components collectively guide the MLLM towards more accurate reasoning and answer generation. Experimental results demonstrate that NoteMR outperforms existing state-of-the-art methods, validating its effectiveness. We hope this work inspires further exploration of integrating knowledge and visual content to advance VQA performance.

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